# Oberwolfach Arbeitsgemeinschaft (ID: 2642)
# New Homological Methods in Toric Geometry

Organizers: Michael K. Brown, David Eisenbud, and Daniel Erman

Sunday 11 October - Friday 16 October 2026

**Schedule of talks:**

| no. | time | title of talk | name |
|---|---|---|---|
| 1 | Mo 9:00-9:50 am | Derived category of projective space, I | Amal Matoo |
| | Mo 10:00-10:30 am | Coffee break | |
| 2 | Mo 10:30-11:20 am | Derived category of projective space, II | Abhik Pal |
| 3 | Mo 11:30-12:20 pm | Derived category of projective space, III | Isidora Bailly-Hall |
| 4 | Mo 3:30-4:20 pm | Toric varieties and Cox rings I | Sharon Robbins |
| | Mo 4:30-5:00 am | Coffee break | |
| 5 | Mo 5:00-5:50 pm | Toric varieties and Cox rings II | Caitlin Davis |
| 6 | Tu 9:00-9:50 am | The Bondal-Thomsen Collection | Andreas Hochenegger |
| | Tu 10:00-10:30 am | Coffee break | |
| 7 | Tu 10:30-11:20 am | Constructing the Hanlon-Hicks-Lazarev resolution | Jason McCullough |
| | Tu 11:30-12:00 am | Discussion: Computation | |
| 8 | Tu 3:30-4:20 pm | More examples and features of the HHL resolution | Ryan Watson |
| | Tu 4:30-5:00 am | Coffee break | |
| 9 | Tu 5:00-5:50 pm | Proof of the HHL Theorem | Greg Smith |
| 10 | We 9:00-9:50 am | Applications of the HHL Theorem | Juliette Bruce |
| | We 10:00-10:30 am | Coffee break | |
| 11 | We 10:30-11:20 pm | The secondary fan | Lauren Cranton Heller |
| 12 | We 11:30-12:20 am | The Cox category, I: Definition | Martin Kalck |
| | We Afternoon | Free for hikes etc. | |
| | We 7:30-8:30 am | Discussion: Toric stacks | |
| 13 | Th 9:00-9:50 am | The Cox category, II: Fourier-Mukai transforms | John Cobb |
| | Th 10:00-10:30 am | Coffee break | |
| 14 | Th 10:30-11:20 am | Proof of revised King's conjecture, I: Outline | Milena Hering |
| | Th 11:30-12:00 am | Open Q&A | |
| 15 | Th 3:30-4:20 am | Proof of revised King's conjecture, II: Exceptionality | Patrick Kennedy-Hunt |
| | Th 4:30-5:00 am | Coffee break | |
| 16 | Th 5:00-5:50 pm | Proof of revised King's conjecture, III: Conclusion | John Nolan |
| 17 | Fri 9:00-9:50 am | Applications of Cox categories | Prashanth Sridhar |
| | Fri 10:00-10:30 am | Coffee break | |
| 18 | Fr 10:30-11:20 am | Tate resolutions over projective space | Matthias Zach |
| 19 | Fr 11:30-12:20 am | Tate resolutions over toric varieties | Sasha Zotine |

Arrival day: Sunday afternoon/evening
Breakfast 8-9 am, lunch 12:30-1:30 pm, dinner 6:30-7:30 pm (In case of a public holiday such as Easter Monday, lunch/dinner take place in Hotel Hirschen and start at 1:15 pm / 6 pm respectively.)
Wednesday afternoon: traditional excursion
Departure day: Friday, before dinner-time